

Can Large Language Models Accurately Generate Answer Keys for Health-related Questions?

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Abstract

The evaluation of text generated by large language models (LLMs) remains a challenge for question answering, retrieval augmented generation (RAG), summarization, and many other natural language processing tasks. Evaluating the factuality of LLM-generated responses is particularly important in medical question answering, where the stakes are high. One method of evaluating the factuality of text is through the use of information nuggets (answer keys). Nuggets are text representing atomic facts that may be used by an assessor to make a binary decision as to whether the fact represented by said nugget is contained in an answer. Although manual nugget extraction is expensive and time-consuming, recent RAG shared task evaluations have explored automating the nuggetization of text with LLMs. In this work, we explore several approaches to nugget generation for medical question answering and evaluate their alignment with expert human nugget generation. We find providing an example and extracting nuggets from an answer to be the best approach to nuggetization. While, overall, we found the capabilities of LLMs to distill atomic facts limited, Llama 3.3 performed the best out of the models we tested.

1 Introduction

Evaluation of automatically generated answers is a major bottleneck in the development of question-answering approaches. Although a need for new evaluation approaches was noticed as soon as the system-generated answers became more complex and abstract (Chen et al., 2019), to date, there are no widely accepted evaluation metrics to approximate human judgments on the quality and other aspects of the generated answers. The TREC 2024 Retrieval Augmented Generation track evaluation (Pradeep et al., 2024) revisited the nugget-based evaluation originally developed for judging answers to definition questions in the 2003 question answering track (Voorhees, 2004). Briefly, a

nugget-based evaluation is a two-step process. In the first step, the assessors create a list of atomic facts (nuggets) that must be present in an answer for the answer to be judged correct and complete. In the second step, the assessors manually map each statement in a system-generated answer to the nuggets. Various performance metrics may be then computed. For example, the TREC 2024 RAG track labeled nuggets as *supported*, *partially supported*, or *not supported* by the answer and then computed the system scores by summing the scores of all nuggets and dividing the sum by the number of nuggets. Most importantly, the evaluation has shown correlations between scores derived from an automatic nugget evaluation and a manual nugget evaluation. This supports the belief that LLMs can be used to support evaluations as long as LLMs are not generating the ground truth (Soboroff, 2024).

To that end, we explore various approaches to the first step of the evaluation - LLMs’ ability to generate atomic factual statements (*cf.* Table 1). This is particularly important in medical question answering, where, although infrequently, as demonstrated by the results of the 2024 TREC BioGen track evaluation, generated answers may contain inappropriate and potentially harmful information (Gupta et al., 2024).

The contributions of this work are as follows:

- We manually generate nuggets for the 2024 BioGen track topics.
- We propose a series of automated nugget-generation approaches considering the question, answer, and relevant documents.
- We evaluated the capabilities of LLMs’ to generate nuggets.

Related work: Nugget generation could be viewed as a form of outline generation in two different settings: 1) a model or a person generating the nuggets has access to a set of answers or docu-

Query: "What will mutation in runx2 affect in the future?"

Answer: "The effect of the runx2 mutation depends on the kind of the mutation. This gene mostly affects bone development. Mutations can cause bone deformities, height lower than expected, extra teeth and other dental problems..."

Manually Extracted Nuggets:

1. Affects (runx2 mutation, bone development)
2. Cause (runx2 mutation, bone deformities)
3. Cause (runx2 mutation, height lower than expected)
4. Cause (runx2 mutation, extra teeth)
5. Cause (runx2 mutation, dental problems) ...

LLM (Llama 3.3) Generated Nuggets:

1. Runx2 mutation affects bone development
2. Runx2 mutation can cause bone deformities
3. Runx2 mutation can result in lower than ...
4. Runx2 mutation can lead to extra teeth
5. Runx2 mutation can cause other dental problems

Table 1: An example of nuggets extracted by a human and LLM for the same query and answer pair.

ments containing information needed to generate the answer; 2) the model or person are provided only with the question and have to generate the outline using their background knowledge. While, to the best of our knowledge, work on direct nugget generation is limited to the above RAG evaluation and an evaluation in which the initial set of test nuggets is generated using ChatGPT (Dietz, 2024; Farzi and Dietz, 2024), the related work on outline generation includes story generation (Wang and Kreminski, 2024) and natural language outline for code generation (Shi et al., 2024). For medical question answering, nugget-based evaluation was revisited in the evaluation of answers to questions about COVID-19 asked by patients and clinicians (Goodwin et al., 2022). The nuggets were generated manually in this evaluation.

2 Methods

2.1 Manual Nugget Generation

For the purpose of having ground truths to evaluate LLM generated nuggets against, we provide expert-curated, manually generated nuggets for the 2024 BioGen track topics. Nuggets were captured from 40 ground truth answers. Each nugget was captured as a semantic triplet in Predicate (subject, object) form. Nuggets were identified by manually

assessing the atomic facts represented in each sentence and their corresponding predicate, subject, and object. Some sentences may contain multiple atomic facts, for example, sentences comprising multiple phrases conjoined by a coordinating conjunction or lists. In such cases, the semantic triple is identified for each phrase or item in the list and recorded separately. Predicates were normalized across the dataset by mapping to a list of expert-curated predicates deemed to be complete in their coverage of the dataset and in conveying represented facts. Each medical concept contained in either the subject or object was associated with a Concept Unique Identifier (CUI) from the Unified Medical Language System (UMLS) (Lindberg et al., 1993). These associations were made by manually assessing the closest match, if any, from the UMLS Metathesaurus Browser. Some facts required more complex nugget structures including, but not limited to, "if, then" clauses and comparisons. These nuggets preserve the underlying logical structure from the answer. We generated a total of 498 nuggets from 40 question-answer pairs which has an average of 12.45 nuggets. Each nugget was reviewed by at least two reviewers.

2.2 LLM-based Nugget Generation

Model Architectures: We tested both popular open-source and proprietary models for nugget generation. The list of models includes Llama (Grattafiori et al., 2024), Gemma (Riviere et al., 2024), Mistral (Jiang et al., 2023), Phi (Abdin et al., 2024), Qwen (Qwen et al., 2025), Vicuna (Chiang et al., 2023), Falcon (Almazrouei et al., 2023), DeepSeek (DeepSeek-AI et al., 2025), and GPT (OpenAI et al., 2024). For some families of models, we included both larger and smaller versions.

Generation Strategies: We developed extensive strategies to generate the nuggets by considering different inputs to the LLMs. Specifically, we used questions, reference answers, and cited documents provided for each assertion in the reference answer. We instructed the models to generate the appropriate nuggets. The detailed strategies are as follows: **(1) Question:** In the first strategy, we only provide the question to the LLMs and instruct them to generate all pertinent nuggets that directly address the user’s query. We started with the zero-shot approach and extended our experiments to the few-shot approach to enable in-context learning, where

we provide an example question and corresponding nuggets in the prompt to direct the model toward better performance. We call this strategy Q_0 (zero-shot) and Q_1 (one-shot).

(2) Question + Answer: We aim to assess LLMs' capability of distilling nuggets from the ground-truth answers. We hypothesized that LLMs are expected to perform well in this setting and it can be considered an upper bound for the first strategy. Similar to the first strategy, we devise two strategies (zero and one-shot) and call them QA_0 and QA_1 .

(3) Question + Documents: Following the success of the retrieval augmented generation (RAG) approach in BioGen (Gupta et al., 2024), we devised another strategy, where the relevant documents along with the question were passed as input to LLMs. To get the relevant documents, we used the two-stage approach, in which we first used BM25 to retrieve the top 100 relevant documents from the BioGen 2024 PubMed corpus, and then re-ranked and selected the top 10 relevant documents using GraphMonoT5 approach (Gupta and Demner-Fushman, 2024). We aimed to investigate the role of input documents in the model's capability of refining the final nuggets. Toward this, we developed two variants of this approach. In the first variant, we feed all the retrieved documents together to the model, and in the second variant, we feed each document sequentially and instruct the model to refine the nuggets and produce the final nuggets at the end of the iteration. We call the former variant QRD_{all} and the latter QRD_{seq} . We also extended this strategy to the ground-truth documents and used the cited documents associated with each assertion in the reference answer. We call these variants QGD_{all} (all documents together) and QGD_{seq} (sequential documents). We have provided all the prompts and experimental details in the **Appendix**.

2.3 Evaluation Metrics

For a given question Q , and its ground-truth nuggets $Y = \{y_1, y_2, \dots, y_m\}$ and model nuggets $X = \{x_1, x_2, \dots, x_n\}$ of the size m and n respectively, we aim to match each nugget $x_i \in X$ to one of the ground-truth nuggets $y_j \in Y$. We formulate the nuggets matching as an assignment problem, where we first compute the semantic similarity $sim(x_i, y_j) = cosine(emb_{x_i}, emb_{y_j})$ between $x_i \in X$ and $y_j \in Y$ and create a similarity matrix $S \in \mathcal{R}^{m \times n}$. We then group all the elements of

matrix S and sort them in descending order. Iteratively, we assign each x_i to y_j if, $S_{ij} \geq \theta$, and x_i and y_j have not been assigned. We continue the process until all x_i (having $sim(.) \geq \theta$) has been assigned. We keep track of each assigned y_j and ensure each y_j is mapped to at most one x_i while maximizing global similarity. Once the assignment is done, we compute precision $p = \frac{|X \cap Y|}{m}$ and recall $r = \frac{|X \cap Y|}{n}$, and F1-score, where $|X \cap Y|$ denotes the number of generated nuggets that match the ground-truth nuggets. For computing semantic similarity, we use the SentenceTransformer (Reimers and Gurevych, 2019) model¹.

3 Results and Discussion

Key Results: Table 2 shows the experimental results of multiple generation strategies at the optimal² value of threshold θ . For the **Question** strategy, the Llama 3.3 (70B) model obtained the maximum F1-score of 34.03% in zero-shot setting. On the **Question + Answer** strategy, which can be considered as an upper-bound for the LLMs, the Llama 3.3 (70B) model achieved a maximum F1-score of 76% in one-shot setting. Under **Question + Document** strategy, all the LLMs exhibited suboptimal performance and showed a maximum of 29.63% F1-score with the GPT-4o model.

Discussion and Findings: We observed a **significant performance gap among the generation strategies**. The best model's performance difference between **Question + Answer** and **Question** is 42.55%. Similarly, we recorded a performance difference between **Question + Answer** and **Question + Document** as 46.37%. With the ground-truth documents as well, the best LLM (GPT-4o) obtained an F1-score of 29.63% compared to its counterpart **Question** strategy with an F1-score of 33.48%. We observed that smaller models (3B-14B) tend to obtain lower performance compared to their counterpart larger models. The study also reveals that **LLMs lacks the capability of accurately generating or extracting nuggets** for the health-related query. Table 2 exhibits the performance of **Question** strategy that tests the LLMs' knowledge in generating the nuggets for the given question, which does not achieve the anticipated performance. For the **Question + Answer** strategy,

¹<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

²The optimal value (0.7) was determined by manual comparison of 10 different sets of LLM and ground-truth nuggets.

Models	Question		Question + Answer		Question + Document			
	Q_0	Q_1	QA_0	QA_1	QGD_{all}	QRD_{all}	QGD_{seq}	QRD_{seq}
DeepSeek-R1 (7B)	17.64	16.57	31.31	43.73	9.05	6.56	10.87	6.83
DeepSeek-R1 (70B)	20.9	22.64	49.34	53.47	12.71	7.87	16.24	12.8
Falcon 3 (7B)	19.02	17.94	48.28	44.25	12.84	9.13	17.64	12.24
Falcon 3 (10B)	22.88	28.21	51.23	60.32	13.06	9.51	18.95	12.56
Gemma 2 (9B)	23.91	23.93	50.4	62.03	11.42	10.88	19.06	17.95
Gemma 2 (27B)	27.11	27.42	59.61	65.16	13.83	9.52	18.42	13.09
Llama 3.2 (3B)	19.15	15.52	36.9	50.32	12.88	11.15	17.23	10.84
Mistral Small (24B)	26.97	28.65	41.02	67.41	12.3	8.64	21.25	17.16
Phi-4 (14B)	26.57	26.32	61.56	66.33	11.71	9.06	26.49	15.24
Qwen2.5 (7B)	22.33	25.96	64.38	65.2	10.96	7.72	20.07	10.94
Qwen2.5 (72B)	28.39	34.52	67.45	72.68	10.71	7.78	29.34	24.39
Vicuna1.5 (7B)	15.77	16.01	44.21	36.44	6.72	4.27	8.09	5.9
Vicuna1.3 (33B)	18.02	22.36	53.35	60.63	8.27	5.12	12.20	7.97
Llama 3.3 (70B)	34.03	33.45	68.32	76	17.14	10.3	29.37	22
GPT-4o	33.48	31.22	64.03	69.82	29.63	20.16	24.54	21.17

Table 2: Performance comparison of various open and closed-source LLMs on the task of nugget generation under different generation strategies. All the results are reported here denote the F1-score.

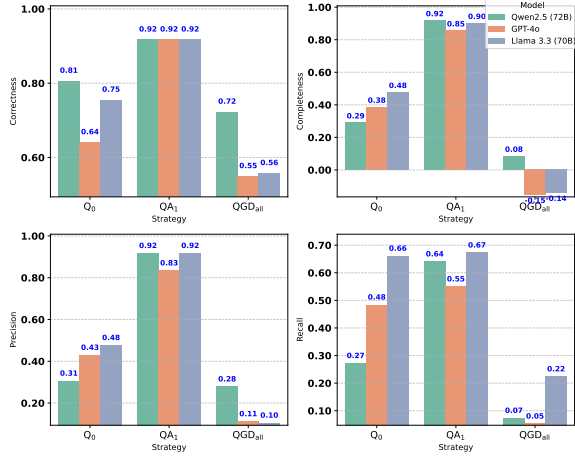


Figure 1: Performance comparison of different models under different generation strategies on multiple human evaluation criteria.

where the ground-truth answer was given to the model to extract the nuggets, it only achieves the F1-score of 76% which highlights LLMs limitation in accurately distilling the atomic facts from the answer. LLMs showed similar behavior when the documents were given to the model for generating/extracting the nuggets.

Human Evaluation: We also performed an extensive human analysis on the LLM-generated nuggets on multiple evaluation criteria. For the human evaluation, we chose the top-3 best-performing LLMs across multiple settings. We evaluated a total of 320 model-generated nuggets with 37 ground-truth nuggets for three diverse questions across three different settings: Q_0 , QA_1 , and

QGD_{all} . We evaluate the quality of each nugget on the following criteria: (a) **Correctness**: whether the generated nugget is correct (2), partially correct (1), and incorrect (0); (b) **Completeness**: whether the generated nugget is misleading (-1), not required (0), Okay, but not required (1), and required (2); (c) **Precision**: portions of the generated nuggets that are correct; (d) **Recall**: portions of the ground-truth nuggets covered in the generated nuggets. We computed all the aforementioned scores for each question and averaged them to report (Fig. 1) the overall scores. The human evaluation of completeness and correctness criteria reveals that under the ground-truth answers (QA_1 strategy) all three LLMs’ performance was better, only the question (Q_0) strategy obtained the sub-optimal performance. The evaluation also highlights that automatic precision and recall are highly aligned with manual precision and recall.

4 Conclusions

This work presented a comprehensive study on generating nuggets for health-related questions using various open and closed-source LLMs. Firstly, we manually formulated nuggets for BioGen 2024 topics and thereafter, we devised multiple nugget generation strategies to assess the capability of LLMs under different settings. We found that most LLMs obtained sub-optimal performance on the task which demonstrates the challenge involved with nugget generation, and we believe that our manual-created nuggets will promote further research in this direction.

Ethics Statement

The health-related questions and reference answers used in this study are publicly available within the Text REtrieval Conference (TREC) 2024 data.

Limitations

While the BioGen 2024 dataset covers a broad range of question topics and intents sourced out of popular health-related searches, it is not exhaustive. Subsequently, our findings on the LLMs' ability to generate nuggets in zero-shot settings apply to information needs covered in the data: clinical decision-support, factoid, and treatment and environment effects. The manual nugget evaluation approach outlined above can be used in the future to expand the data.

Another limitation of the data is a single reference answer. While the bulk of the nuggets must be present in any answer, some of the automatically generated nuggets could have been present in alternative answers. For example, if the existing reference answer list surgery as a treatment option, without specifying the best procedures, automatically generated nuggets that name specific surgeries will not get any credits for these nuggets. While this may somewhat lower the scores, it should not affect the model ranking, as the same approach is used for all models. In the future, more than one reference answer would be desirable to base the evaluation on a nugget pyramid (Marton and Radul, 2006).

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A Models

Table 3 is an exhaustive list of the models tested in our experiments with their versions and approximate number of parameters.

Model	Version / Size
Llama 3.2	3B ³
Llama 3.3	70B ⁴
Gemma 2	9B ⁵ , 27B ⁶
Mistral Small	24B ⁷
Phi-4	14B ⁸
Qwen2.5	7B ⁹ , 72B ¹⁰
Vicuna _{1.5}	7B ¹¹
Vicuna _{1.3}	33B ¹²
Falcon 3	7B ¹³ , 10B ¹⁴
DeepSeek-R1	7B ¹⁵ , 70B ¹⁶
GPT-4o	gpt-4o-2024-08-06

Table 3: A list of the models tested in our experiments.

B Details of Experimental Setup

Details of LLM Prompting To test the capabilities of LLMs to generate and extract nuggets, we prompted each model with a series of approaches.

³<https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct>

⁴<https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct>

⁵<https://huggingface.co/google/gemma-2-9b-it>

⁶<https://huggingface.co/google/gemma-2-27b-it>

⁷<https://huggingface.co/mistralai/Mistral-Small-24B-Instruct-2501>

⁸<https://huggingface.co/microsoft/phi-4>

⁹<https://huggingface.co/Qwen/Qwen2.5-7B-Instruct>

¹⁰<https://huggingface.co/Qwen/Qwen2.5-72B-Instruct>

¹¹<https://huggingface.co/lmsys/vicuna-7b-v1.5>

¹²<https://huggingface.co/lmsys/vicuna-33b-v1.3>

¹³<https://huggingface.co/tiiuae/Falcon3-7B-Instruct>

¹⁴<https://huggingface.co/tiiuae/Falcon3-10B-Instruct>

¹⁵<https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Qwen-7B>

¹⁶<https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Llama-70B>

Table 4 contains the prompt used for Q_0 , the zero-shot variation of our **Question** strategy. Table 5 contains the prompt used for QA_0 , the zero-shot variation of our **Question + Answer** strategy. Table 6 contains the prompt used for Q_1 , the one-shot variation of our **Question** strategy. Table 7 contains the prompt for QA_1 , the one-shot variation of our **Question + Answer** strategy. Table 8 contains the prompt for QGD_{all} and QRD_{all} , our **Question + Document** strategies with all ground-truth documents and all retrieved documents, respectively. Table 9 contains the prompt for QRD_{seq} and QGD_{seq} , our **Question + Document** strategies with sequential ground-truth documents and sequentially retrieved documents, respectively.

Each prompt contains some combination of query (q), answer (a), context (c), and initial nugget list (i) variables. The query and answer variables were substituted with each query and answer from the BioGen 2024 dataset. For the settings with all documents, the context variable was substituted with a list of the abstracts from all documents for the query separated by a new line character. For the settings with sequential documents, the context variable was substituted with a single abstract. The prompt for the sequential documents settings also contained the initial nugget list variable. This variable was initially "None" and then was substituted with the list of nuggets produced by the model provided with the previous abstract. For the sequential documents settings, the models were prompted once with each abstract and only the final list of nuggets was recorded. All models were prompted with their default settings (e.g. temperature).

SYSTEM: You are NuggetGenerateLLM, an AI assistant specialized in generating all information nuggets that are required to completely answer a given query. A nugget is an atomic fact. USER: Generate all the information nuggets that are required to completely answer the query given below. Each nugget must contain one, and only one, fact. A nugget must be as concise and as specific as possible. A nugget cannot contain a list, each element in a list must be its own nugget. Each nugget must directly answer the query. The list of nuggets must not contain redundant information. Return a list of nuggets such that each nugget is on a new line. Do not number or bullet the list. Do not include anything in your response except for the list of nuggets. Here is an example of the output format: <pre>nugget1 nugget2 ... Query: q LLM: nugget1 nugget2 ...</pre>
--

Table 4: Prompt for Q_0 .

SYSTEM: You are NuggetExtractLLM, an AI assistant specialized in extracting information nuggets from a given answer. A nugget is an atomic fact. USER: Generate all the information nuggets that are required to completely answer the query given below. Each nugget must contain one, and only one, fact. A nugget must be as concise and as specific as possible. A nugget cannot contain a list, each element in a list must be its own nugget. Each nugget must directly answer the query. The list of nuggets must not contain redundant information. Return a list of nuggets such that each nugget is on a new line. Do not number or bullet the list. Do not include anything in your response except for the list of nuggets. Here is an example of the output format: <pre>nugget1 nugget2 ... Query: q Answer: a LLM: nugget1 nugget2 ...</pre>
--

Table 5: Prompt for QA_0 .

SYSTEM: You are NuggetGenerateLLM, an AI assistant specialized in generating all information nuggets that are required to completely answer a given query. A nugget is an atomic fact.

USER: Generate all the information nuggets that are required to completely answer the query given below. Each nugget must contain one, and only one, fact. A nugget must be as concise and as specific as possible. A nugget cannot contain a list, each element in a list must be its own nugget. Each nugget must directly answer the query. The list of nuggets must not contain redundant information. Return a list of nuggets such that each nugget is on a new line. Do not number or bullet the list. Do not include anything in your response except for the list of nuggets. Here is an example of the output format:

nugget1
nugget2
...

Here is an example query: Why is transferrin and iron low in covid patients but ferritin high?
This is the list of nuggets that should be generated for this query:

Lymphocytes and viruses compete for iron.
Lymphocytes need iron for cellular response.
Lymphocytes need iron for humoral response.
Viruses need iron to replicate.
Infection lowers iron levels in the blood.
Infection increases ferritin levels in the blood.
High ferritin is associated with increased mortality.
Iron homeostasis needs ferritin.
Ferritin is involved in physiologic processes.
Ferritin is involved in pathologic processes.
High ferritin indicates response to inflammation.
High ferritin levels are linked to poor outcomes of COVID-19.
Iron depletion therapy showed anti-viral activity in the COVID-19 pandemic.
Iron depletion therapy showed anti-fibrotic activity in the COVID-19 pandemic.

Query: q

LLM:
nugget1
nugget2
...

Table 6: Prompt for Q_1 .

SYSTEM: You are NuggetExtractLLM, an AI assistant specialized in extracting information nuggets from a given answer. A nugget is an atomic fact.

USER: List all of the information nuggets in the answer given below that are required to completely answer the query. Each nugget must contain one, and only one, fact from the answer. A nugget must be as concise and as specific as possible. Each element in a list must be its own nugget. Each nugget must directly answer the query. The list of nuggets must not contain redundant information. Return a list of nuggets such that each nugget is on a new line. Do not number or bullet the list. Do not include anything in your response except for the list of nuggets. Here is an example of the output format:

nugget1
nugget2
...

Here is an example query: Why is transferrin and iron low in covid patients but ferritin high?
This is the list of nuggets that should be generated for this query:

Lymphocytes and viruses compete for iron.
Lymphocytes need iron for cellular response.
Lymphocytes need iron for humoral response.
Viruses need iron to replicate.
Infection lowers iron levels in the blood.
Infection increases ferritin levels in the blood.
High ferritin is associated with increased mortality.
Iron homeostasis needs ferritin.
Ferritin is involved in physiologic processes.
Ferritin is involved in pathologic processes.
High ferritin indicates response to inflammation.
High ferritin levels are linked to poor outcomes of COVID-19.
Iron depletion therapy showed anti-viral activity in the COVID-19 pandemic.
Iron depletion therapy showed anti-fibrotic activity in the COVID-19 pandemic.

Query: q
Answer: a

LLM:
nugget1
nugget2
...

Table 7: Prompt for QA_1 .

SYSTEM: You are NuggetGenerateLLM, an AI assistant specialized in using context to generate all information nuggets that are required to completely answer a given query. A nugget is an atomic fact.

USER: Use the context provided to generate all the information nuggets that are required to completely answer the query given below. Each nugget must contain one, and only one, fact. A nugget must be as concise and as specific as possible. A nugget cannot contain a list, each element in a list must be its own nugget. Each nugget must directly answer the query. The list of nuggets must not contain redundant information. Return a list of nuggets such that each nugget is on a new line. Do not number or bullet the list. Do not include anything in your response except for the list of nuggets. Here is an example of the output format:

nugget1
nugget2
...
Query: q
Context: c
LLM:
nugget1
nugget2
...

Table 8: Prompt for QGD_{all} and QRD_{all} .

SYSTEM: You are NuggetGenerateLLM, an AI assistant specialized in using context to update a list of all information nuggets that are required to completely answer a given query. A nugget is an atomic fact.

USER: Use the context provided to update the list of information nuggets, if needed. The list should contain all nuggets that are required to completely answer the query given below. If no list of nuggets is provided, generate a list of nuggets. Each nugget must contain one, and only one, fact. A nugget must be as concise and as specific as possible. A nugget cannot contain a list, each element in a list must be its own nugget. Each nugget must directly answer the query. The list of nuggets must not contain redundant information. Return a list of nuggets such that each nugget is on a new line. Do not number or bullet the list. Do not include anything in your response except for the list of nuggets. Here is an example of the output format:

nugget1
nugget2
...
Query: q
Context: c
Initial Nugget List: i
LLM:
nugget1
nugget2
...

Table 9: Prompt for QGD_{seq} and QRD_{seq} .